

Problem

A balanced three-phase source furnishes power to the following three loads:

Load 1: 6 kVA at 0.83 pf lagging

Load 2: unknown

Load 3: 8 kW at 0.7071 pf leading

If the line current is 84.6 A rms, the line voltage at the load is 208 V rms, and the combined load has a 0.8 pf lagging, determine the unknown load.

Step-by-step solution

Step 1 of 2

$$\begin{aligned}
 S &= S_1 + S_2 + S_3 \\
 &= 6[0.83 + j \sin(\cos^{-1} 0.83)] + S_2 + 8(0.7071 - j0.707) \\
 &= 10.6368 - j2.31 + S_2 \text{ KVA} \rightarrow (1)
 \end{aligned}$$

But,

$$\begin{aligned}
 S &= \sqrt{3} V_L I_L \angle \theta = \sqrt{3} (208)(84.6)(0.8 + j0.6) \text{ VA} \\
 &= 24.383 + j18.287 \text{ KVA} \rightarrow (2)
 \end{aligned}$$

Step 2 of 2

For (1) & (2),

$$S_2 = 13.746 + j20.6 = 24.76 \angle 56.28 \text{ KVA}$$

Thus, the unknown load is,

$$24.76 \text{ KVA at } 0.8851 \text{ P f Lagging} .$$